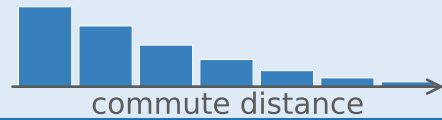


## ① Decay Calibration

### SOURCE CITIES

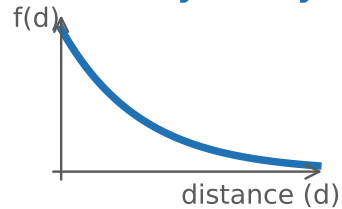


### Aggregate Distance Distribution



### Tanner model

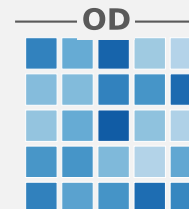
### Mobility Decay



transferred  
decay

### Predicted OD Matrix

$$flow(i \rightarrow j) = O_i \times A_j \times f(\text{distance}_{ij})$$



## ② Target-City Estimation

### TARGET CITY



### Attraction Score



### Outflow Model



## ③ Performance Decomposition

### What limits accuracy?

Decay transfer  $\approx$  solved

Oracle Outflow

CPC 0.743

Local Calibration

CPC 0.744

Outflow = bottleneck

Survey-free

CPC 0.697

Where outflow fails

Dense polycentric metros

$\text{corr}(\text{outflow}, \text{CPC}) = 0.49$

4.7 pp

- Decay transfer (near-lossless)
- OSM open features
- Bottleneck: outflow estimation

Aggregate distance distributions are near-sufficient for mobility-decay calibration.

*The bottleneck is trip-production estimation — not decay transfer.*